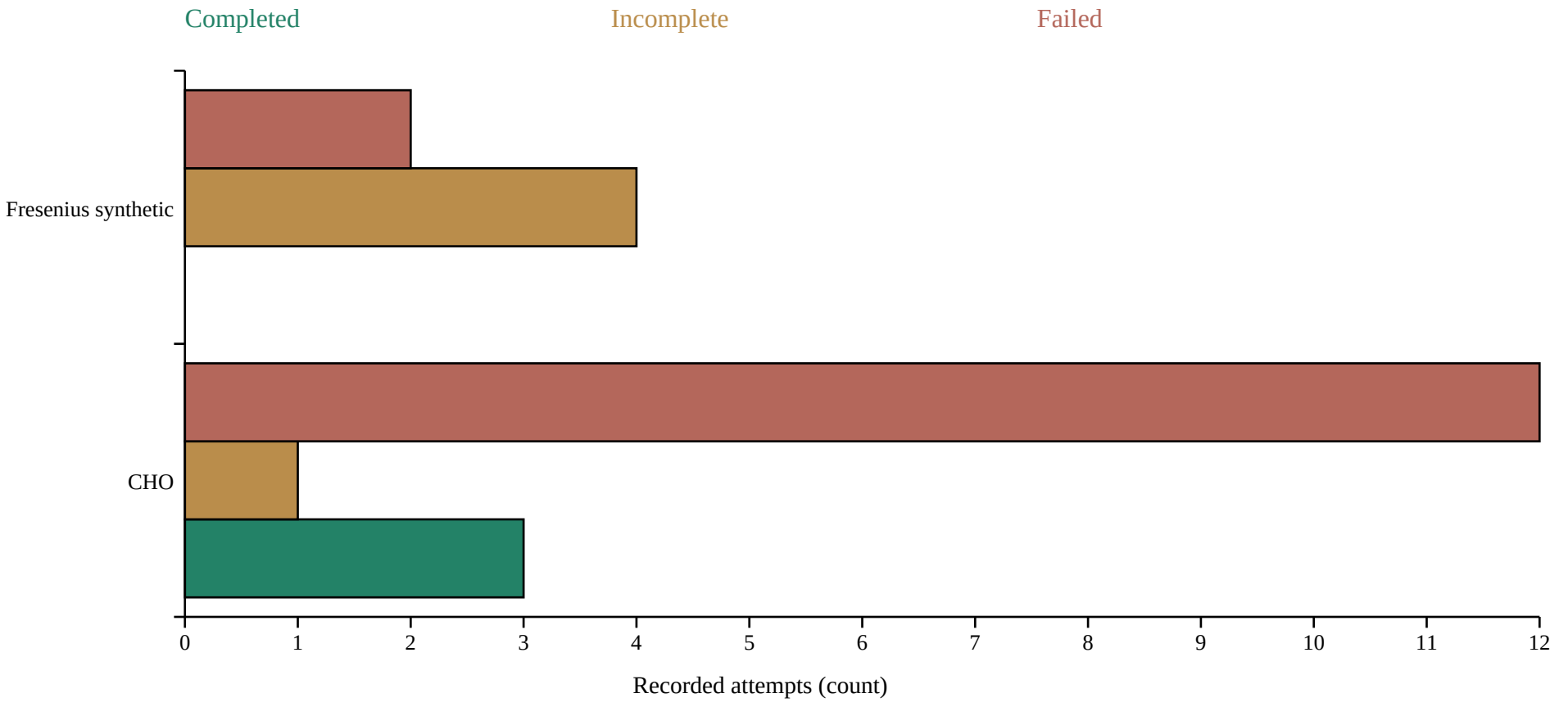
Manufacturing OS: every recorded execution counts



Historical versions and infrastructure failures included. Completion is not scientific qualification.

1 trace-only record(s): method/execution unknown; excluded from known-method denominators.

Fresenius: synthetic manufacturing reconstruction informed by publicly available FDA-related materials.

No internal company data, endorsement or production improvement demonstrated. All publication gates pending.